The viscous inversion of the corner-flow blowup geometry fires at the collapse transient's turnaround, at every viscosity tested

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Abstract

Paper [P1] measured σ_Λ, the scaling exponent of the scale-invariant direction-regularity observable $\Lambda = \sup_P |\nabla\xi|\,|\omega|^{-1/2}$, on the Boussinesq corner-flow blowup scenario: $+1.00 \pm 0.03$ inviscid (calibrated where blowup is proven), inverting to a deep-collapse value near -1.2 under viscosity at $\nu \in \{10^{-4}, 10^{-3}\}$. This paper asks the questions that result forces: how the inversion depends on viscosity, where in the collapse it happens, and what sets that location. Within the tested scenario family (Section 1): (1) *No resolved viscosity dependence.* At the five tested viscosities $\nu \in \{10^{-5}, 3\times10^{-5}, 10^{-4}, 3\times10^{-4}, 10^{-3}\}$ the deep-collapse exponent shows no trend (chord estimator -1.12 to -1.29, every cross-grid spread under the pre-registered 0.15 bar; two-slope-form asymptote -1.07 ± 0.03); the two previously certified viscosities replicated 4/4 against independent runs. Against the window-matched inviscid anchor ($+0.589/+0.585$) the data show no approach toward the inviscid value down to $\nu = 10^{-5}$: any crossover in ν lies below the tested range. The data are consistent with a $\nu \to 0^+$ discontinuity of the exponent; they cannot distinguish it from a crossover below 10^{-5}. (2) *A fixed trigger.* The switch sits at $A^* = 52.7$–$54.2 \times A_0$ in twelve measurements at $\nu \in \{10^{-4}, 10^{-3}\} \times A_0 \in \{50, 100, 200\}$, both grids, and is bracketed inside the same band at the three other viscosities ($A_0 = 100$ only). Since the initial corner-gradient amplitude is $37.7 \times A_0$ (a geometric constant of this IC family), the trigger is a growth factor 1.43 (spread 1.40–1.44) above the initial value: about a third of an e-fold. The switch is nearly a step, completing within ∼0.2–0.3 e-folds (measured at $\nu = 10^{-4}$ only). (3) *The two instruments meet.* An empirical clock fit ($\gamma_A = 0.77$–1.11) places the switch at $s^* = 0.52 \pm 0.05$ in rescaled time (per-run degeneracy widens this to $[0.29, 0.70]$), consistent with the corner-frame march's fast-transient turnaround at $s = 0.5$ (resolved to the $ds = 0.25$ step); the pre-registered lock-on hypothesis ($s \sim 3.7$–13.5) is refuted by more than a factor of five. A large-deviation march closes the linearity gap as far as the march reaches: the $s = 0.5$ turnaround is amplitude-invariant across 4.5 decades of perturbation amplitude, and the march's basin-or-instrument edge lies between amplitudes 3×10^{-2} and 10^{-1} along the one direction tested. The licensed mechanism statement, exactly bounded: viscosity is necessary for the inversion (inviscid runs never invert), and at the two viscosities where the clock was fit the inversion occurs at a rescaled time consistent with the transient's turnaround, well before self-similar settling; the switch location tracks a ν-independent, A_0-linear stage of the collapse whose dynamical identity remains open.

*Sequel to [P1]. Every number regenerates from bytes on disk (Section 9). This paper was drafted after, not before, a two-lens verification pass over its own claim records: roughly 230 numeric claims recomputed independently, twelve scoping corrections applied, and four of our own estimators and readings killed en route. The corpses are in Section 8.

1 Scope, stated first

Everything in this paper is measured on one scenario family: the axisymmetric Boussinesq corner flow of the Luo–Hou type, quartic initial condition, corner scale $r_0 = 0.4$, `zpow` 1, free-slip viscosity in 5D-Laplacian form (the no-slip results of Barker–Prange [7] concern a different boundary condition and are cited as prior art for the criteria, not as the same setting). Amplitudes are in code units of the vector-$|\omega|$ maximum; grids are 128×384 and 256×768 (one refinement doubling; the 0.15 cross-grid bar tests consistency under that doubling, not asymptotic grid convergence); the march runs seed-0 perturbations at $ds = 0.25$ with the M2 stop rule, and every accepted march step converged the true nonlinear residual below 10^{-10}.

Novelty scoping. The prior-art search of [P1] (a one-session web sweep; "not found by this search," never "proven absent") covers the viscosity-resolved exponent curve: no measurement of a direction-regularity exponent as a function of viscosity was found, and Section 2 is that curve. The trigger invariant, switch width, clock comparison, and basin bound are reported as measurements with no novelty claim attached. Adjacent art sighted in a supplementary search: analytical viscous geometric depletion [8, 9], and nearly self-similar viscous blowup in *generalized* axisymmetric Navier–Stokes with dimension as a parameter [10]; our measurements concern the standard equations at fixed dimension, on this scenario family, at the viscosities stated.

2 The curve: no resolved ν-dependence

Ten new DNS runs (five viscosities, both grids, doubled snapshot cadence, fresh tags) went through the certified estimator of [P1] verbatim: signed circulation-drift gate, peak-box Λ recipe, cross-grid common-amplitude overlap, top half, symmetric midpoint cut, OLS slope with 10^4-resample bootstrap intervals.

ν	128×384	256×768	spread
0 (anchor)	$+0.589$	$+0.585$	0.005
10^{-5}	-1.174	-1.190	0.016
3×10^{-5}	-1.271	-1.156	0.115
10^{-4}	-1.119	-1.187	0.068
3×10^{-4}	-1.286	-1.204	0.082
10^{-3}	-1.207	-1.231	0.024

Every spread passes the pre-registered 0.15 bar. The two viscosities certified in [P1] replicated 4/4 against these independent runs (worst $|\Delta| = 0.055$, within tolerance); the three new viscosities are single run-pairs with cross-grid agreement as their check. The deep windows carry $n = 20$–27 snapshots (the doubled cadence fixed [P1]'s thinnest tables). A two-slope-form fit (Section 3) puts the asymptotic deep slope at -1.07 ± 0.03 across all six fitted runs, systematically shallower than the chord values; both are quoted and neither replaces the other. The chord-vs-form gap ($\sim$0.1) is the current estimator systematic on the exponent's absolute value; the ν-flatness conclusion is insensitive to it.

What the table does not show: any approach toward the inviscid anchor. If $\sigma_\Lambda^{\text{deep}}(\nu)$ returns to $+0.59$ as $\nu \to 0$, it does so below $\nu = 10^{-5}$. Within the tested range the inversion is viscosity-blind. The Corollary-0 reading of [P1] (the viscous cap on direction-gradient growth exists for every $\nu > 0$ and is absent at $\nu = 0$ [3]) is consistent with a genuine 0^+ discontinuity; that reading is interpretation, and the measurement alone cannot exclude a crossover below the tested range.

3 The trigger: a fixed growth factor, and a near-step switch

At finite ν, $\sigma(A)$ is not one number: every viscous run crosses from inviscid-like slope (+0.39 to +0.81) at low amplitude to the inverted value at high amplitude. [P1] adjudicated window-averaged slopes as regime mixtures; here the crossover itself becomes the observable.

The estimator lineage is part of the record. A pre-registered chord-intersection estimator for the crossover amplitude *failed* its split-invariance test (15–42% drift tracking the window split, monotone in nine of ten runs) and its outputs are void: $\sigma(A)$ curves smoothly rather than breaking, so a two-chord intersection follows the split point. Its successor, the first zero crossing of a k-row sliding-window local slope, passes the same class of test that killed its predecessor: k-drift 0.9–2.1% on the discovery runs and 0.7–1.0% on the cross-term runs, cross-grid agreement under 1%.

By that estimator, the crossover amplitude is A_0-linear with measured power +1.00 (ratios 0.50/0.49 at $A_0 = 50$ and 2.01/2.00 at $A_0 = 200$ against the amplitude-symmetry prediction), and the twelve measurements at $\nu \in \{10^{-4}, 10^{-3}\} \times A_0 \in \{50, 100, 200\}$ give $A^*/A_0 = 52.7$–54.2. At $\nu \in \{3 \times 10^{-4}, 3 \times 10^{-5}, 10^{-5}\}$ ($A_0 = 100$ only) the crossover is bracketed inside the same fixed band: the lower half of every overlap window reads inviscid-positive and the upper half reads inverted, so the crossover cannot have moved with ν within the tested window. The tested region of the (ν, A_0) plane is a cross, not a tile; the plane statement is exact for that cross.

The initial corner-gradient amplitude is $37.7 \times A_0$ in every run, a geometric constant of the quartic IC family (early inviscid snapshots grow $37.71 \to 37.80 \to 37.93$: the value at the first trusted snapshot is essentially the initial value). The trigger is therefore a growth factor of 1.43 above the initial amplitude (twelve-run spread 1.40–1.44; the tanh-form estimator places the crossing systematically $\sim$8% lower, so the absolute factor is estimator-conditional at that level while A_0-linearity and ν-independence are estimator-robust). Three dependencies are declared: the factor is a property of this IC family, of the vector-$|\omega|$ observable, and of the zero-crossing estimator.

The switch is nearly a step. A two-slope blend fit is residual-adequate on all six runs at $\nu = 10^{-4}$ but parameter-degenerate (the optimizer drives it into an exponential-transient limit; its center, width, and low-A slope are not individually physical and are not quoted). The robust extract is the transient decay scale: the local slope completes its transition from 0 to the inverted value within $\sim$0.2–0.3 e-folds of amplitude. The width is measured at $\nu = 10^{-4}$ only.

One confound was tested and killed before these claims were written: across the switch, the $|\omega|$ argmax stays at the corner ($z^* = 0$, $r^* = 1.000$) at every gated snapshot, and the observable tracks the swirl-gradient structure throughout (ω_θ vanishes at the corner by symmetry). The diagnostic was run at $\nu = 10^{-4}$ on the 128-grid and on the matched inviscid run; we treat it as excluding a structure handoff for the runs where the switch is certified. The inversion is a change in the direction-field geometry at one fixed structure, not the instrument changing targets.

4 The clock: two instruments meet at the transient's turnaround

[P1]'s rescaled-frame march certified the corner profile as an attractor: perturbations peak at $s = 0.5$ (raw factor 2.389 at linear amplitude) and contract at 0.270 per s-unit. The DNS switch can be placed on the same s-axis by fitting the collapse clock $\ln A = -\gamma_A \ln(1 - t/T^*) + \ln A_i$ per run, T^* scanned, with $s = -\ln(1 - t/T^*)$.

run	γ_A	s^*	s^* over 2×-residual T^* set
OR 128×384 ($\nu = 0$)	0.77	0.61	$[0.42, 0.70]$
OR 256×768 ($\nu = 0$)	0.94	0.54	$[0.41, 0.58]$
C 128×384 (10^{-4})	0.86	0.55	$[0.38, 0.61]$
C 256×768 (10^{-4})	1.11	0.48	$[0.35, 0.51]$
C 128×384 (10^{-3})	1.04	0.47	$[0.30, 0.53]$
C 256×768 (10^{-3})	1.03	0.49	$[0.31, 0.55]$

$s^* = 0.52 \pm 0.05$ (run-to-run scatter; the T^*–γ_A degeneracy widens the honest per-run interval to $[0.29, 0.70]$), consistent with the march turnaround at $s = 0.5$, itself resolved only to the $ds = 0.25$ step. The value is conditional on the clock form and on the $54 \times A_0$ switch input: the tanh-implied 49–50 $\times$ A_0 shifts s^* by about -0.09, inside the degeneracy band. γ_A spans 0.77–1.11, consistent with the corner frame's one-e-folding-per-s-unit and measured rather than assumed. The pre-registered hypothesis space put lock-on at $s \sim 3.7$–13.5; even the widest degeneracy bound clears that window by more than a factor of five. Within the pre-registered space and clock family, the lock-on hypothesis is refuted: the switch does not wait for the collapse to settle onto the profile.

The comparison identifies matching timescales of one operator, not coincident trajectories: the march transient is measured for perturbations about the settled profile, while the DNS approach is a large deviation. The next section closes that gap as far as the march can reach.

5 The large-deviation march

The certified march ran the same seed-0 perturbation at amplitudes 3×10^{-3}, 10^{-2}, 3×10^{-2}, and 10^{-1} (the largest previously validated amplitude was 10^{-3}). At the first three, all 240 steps are valid and the raw-norm peak sits at $s_{\text{peak}} = 0.5$ exactly, at step resolution. Combined with [P1]'s M2 ladder, the transient turnaround is amplitude-invariant across 4.5 decades of relative deviation (10^{-6} to 3×10^{-2}). The $s = 0.5$ timescale that the DNS switch lands on is a property of the operator's nonlinear neighborhood, not merely of its linearization.

Around the pinned turnaround, genuine nonlinear structure appears for the first time in this campaign: the transient's end stretches ($s_{\text{transient}} = 13.5 \to 14.0 \to 14.75$, +9%) and the peak factor grows ($2.4068 \to 2.4504 \to 2.5884$ within this ladder; M2's linear value is 2.389) while the peak position does not move. The Lyapunov P-norm contracts monotonically from $V_0 = 77$ through 13.7 decades with zero violations above the pre-registered V-floor artifact threshold: the certificate's contraction holds at 3% deviation.

At amplitude 10^{-1} the march stalls at step one (raw growth 3.3× in one step; the P-norm nevertheless decreased on that step, a mixed signal). The pre-registered protocol does not separate dynamical escape from solver stall, so the honest statement is a basin-or-instrument edge: along the seed-0 direction, $3 \times 10^{-2} < r < 10^{-1}$ in the march's norm. This also bounds the remaining daylight in Section 4's comparison: the DNS initial deviation exceeds this basin, and its approach enters the basin during the transient. Full closure would need DNS-state initialization of the march, which is a different instrument build.

6 Mechanism, stated within bounds

For every viscosity tested, the inversion is present only with $\nu > 0$: inviscid runs on identical amplitude windows never invert. At the two viscosities where the clock was fit, the switch occurs at a rescaled time consistent with the corner-frame transient's turnaround, well before self-similar settling, and the turnaround timescale is amplitude-robust across every amplitude the march can hold. Viscosity is necessary for the inversion; the location of the switch tracks a ν-independent, A_0-linear stage of the collapse. What sets that stage dynamically is open, and Section 2's gap below $\nu = 10^{-5}$ is open with it. The strongest reading consistent with all measurements: the geometric depletion that [P1] measured deep in the collapse is already fully switched on by the end of the approach transient, at every viscosity and amplitude tested, and the deep-collapse exponent it switches to does not depend on how much viscosity did the switching.

7 Relation to prior art

The criteria territory is [P1]'s: Constantin–Fefferman [2], Beirão da Veiga–Berselli [4, 5] at the definition of Λ, Giga–Miura [6] and Barker–Prange [7] for qualitative type-I exclusion under direction coherence, and Constantin's identity [3] behind the Corollary-0 diagnostic. Analytical viscous geometric depletion is developed in [8] and, through logarithmically weighted direction classes, in [9]; neither defines a measured exponent or an onset. Nearly self-similar viscous blowup is reported in *generalized* axisymmetric Navier–Stokes with dimension as a parameter [10]; the present measurements concern the standard equations on this scenario family and are not in tension with results for modified systems. The measured objects of this paper (an exponent-vs-viscosity curve, a trigger invariant, a cross-instrument clock comparison) were not found in the [P1] search; only the curve carries a novelty claim, under that search's stated horizon.

8 What died on the way

Four of our own results were killed by our own instruments during this campaign, and the corpses are retained:

- The chord-intersection crossover estimator: failed split-invariance (15–42% drift); all its outputs, including a fitted crossover-vs-viscosity slope, are void.
- The tanh-form parameters: residual-adequate fit, degenerate parameters; only the decay scale and asymptote survive.
- The "four e-folds of dynamical age" reading of the trigger: killed by the normalization audit (the initial corner-gradient amplitude is $37.7 \times A_0$, not A_0); the honest trigger is $1.43\times$ above the initial value, essentially at collapse onset.
- The lock-on bridge hypothesis: the clocks were compared and it lost, by a factor of five beyond the widest degeneracy bound.

[P1]'s standing retractions (window-averaged viscous slopes; the amplitude-composite reading of the full-window inviscid calibration) are inherited: no full-window σ is quoted anywhere in this paper, and every viscous contrast is drawn against the window-matched inviscid +0.589, never against +1.00.

9 Reproducibility

The DNS campaign is 18 runs (about 90 minutes of laptop wall time) plus the march ladders (∼90 seconds each), all regenerable: run scripts, pre-registration (every estimator, bar, and hypothesis space in this paper was written down before the data it judges existed; the spec file's section dates interleave with the run logs), extraction code, and per-claim records with their adjudication addenda are in the repository alongside [P1]'s verification suite (15/15 standing, watcher live). The two-lens audit that preceded this draft (a numbers tracer recomputing ∼230 claims from the raw arrays and snapshots; an adversarial referee producing the twelve scoping corrections now baked into the text) is itself part of the record.

References

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